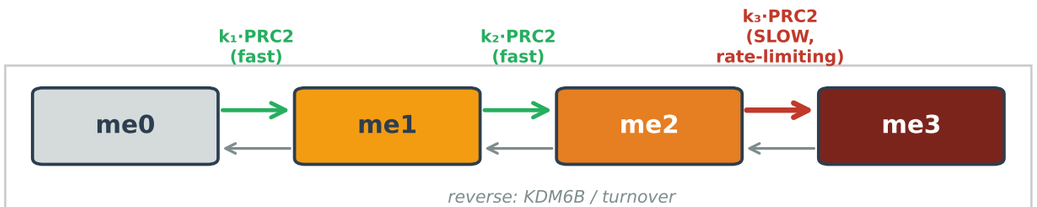
Serial H3K27 methylation (me1→me2→me3) vs a lumped mark — how the slow last step changes me3 kinetics

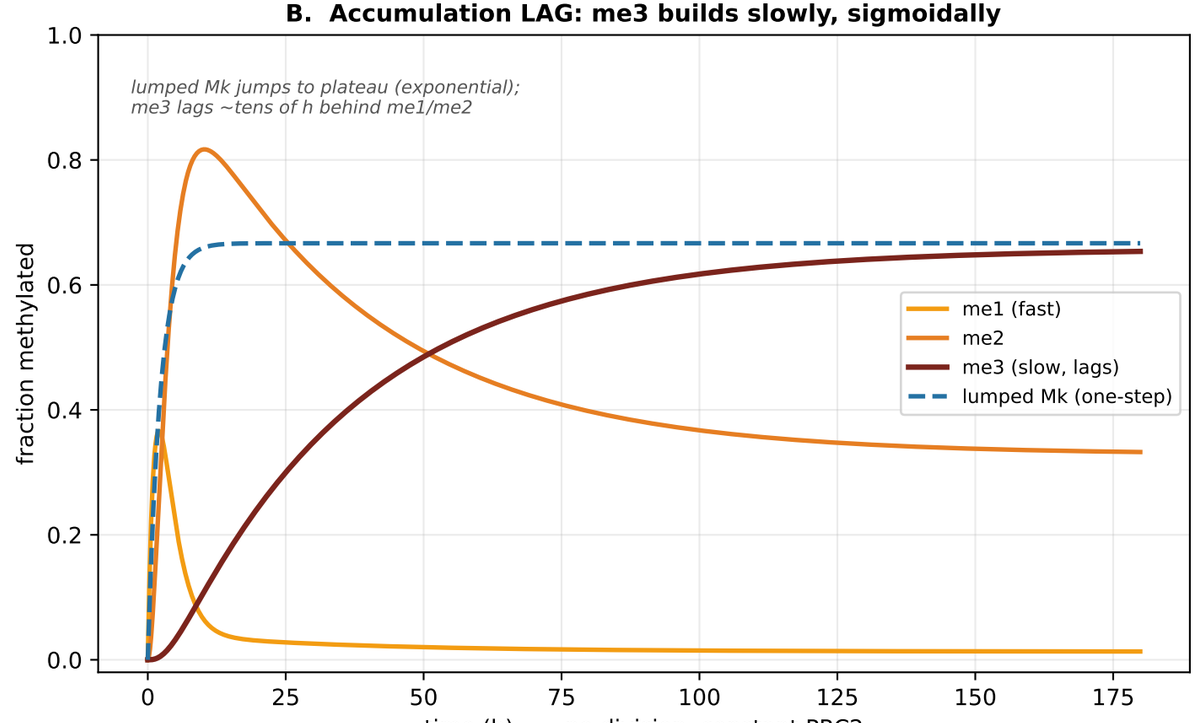
me0→me1→me2→me3 with FAST me1/me2 and a SLOW, rate-limiting me2→me3. Every division halves the modified fractions (new histones = me0). Serial me1/me2/me3 chain (now the default).

A. The serial scheme

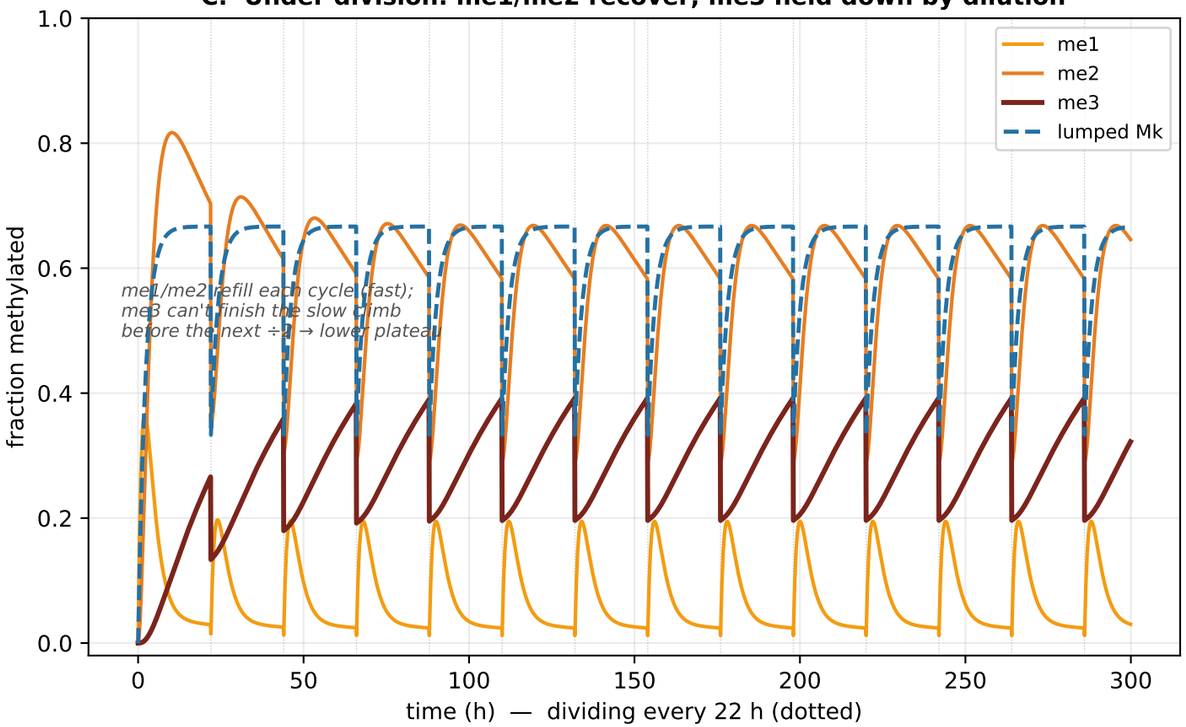


At each division (S phase): new histones are unmodified →
$m_1, m_2, m_3 \rightarrow \frac{1}{2}\cdot m_1, \frac{1}{2}\cdot m_2, \frac{1}{2}\cdot m_3$ (the lumped model's Mk→½Mk)
Why me3 is special:
- me1, me2 equilibrate fast (~2 h) → track current PRC2 activity, high turnover.
- me3 sits behind the slow k_3 bottleneck → builds over ~tens of hours, integrates activity over time → the heritable 'memory' state.

B. Accumulation LAG: me3 builds slowly, sigmoidally



C. Under division: me1/me2 recover; me3 held down by dilution



D. me3 depends strongly on cycle speed — the lumped mark does not

